right action of G on S : $S \times G \to S$ s.t $s \triangleleft e = s$

$(s, g) \mapsto s \triangleleft g$ $\quad (s \triangleleft g) \triangleleft h = s \triangleleft (gh)$

linear action:

$(\alpha v + \beta w) \triangleleft g$

$= \alpha (v \triangleleft g) + \beta (w \triangleleft g)$

- right action of G on P | $u \in P$ $\quad u \triangleleft g \in P$
- right action of G on $\mathfrak{g}$ | $A \in \mathfrak{g}$ $\quad A \triangleleft g = g^{-1} A g = Ad_{g^{-1}}(A)$

$(A \triangleleft g) \triangleleft h = (g^{-1} A g) \triangleleft h = h^{-1} g^{-1} A g h$

$= (gh)^{-1} A g h = A \triangleleft gh$

- right action of G on $\mathfrak{X}(P)$ (set of vector field on P)

↳ section of TP

that is a map $X : P \to TP$ $\quad X_u \in T_u P$

$u \mapsto (u, X_u)$

$X \circ \pi = id$

$R_g : P \to P$ right multiplication

$u \in P$ $\quad u \triangleleft g = R_g(u)$

$(dR_g)_u : T_u P \to T_{u \triangleleft g} P$ $\quad (f : M \to N \quad df_a : T_a M \to T_{f(a)} N$

and since R_g is invertible $(R_g^{-1} = R_{g^{-1}})$ $\quad (dR_g)_u$ also invertible

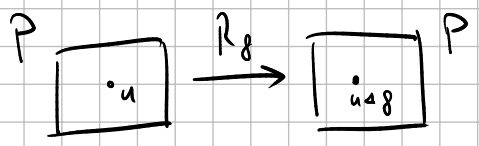
to prove use action of derivation on functions

$(dR_g)_u^{-1} = (dR_{g^{-1}})_{u \triangleleft g}$

allows us to define $X \triangleleft g$ if X is a vector field

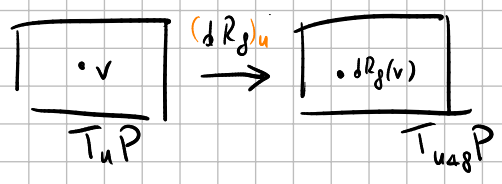
$(X \triangleleft g)_u = (dR_g)_u^{-1} (X_{u \triangleleft g})$

$= (dR_{g^{-1}})_{u \triangleleft g} (X_{u \triangleleft g})$



prove that it is a right action

$(X \triangleleft g \triangleleft h)_u = (X \triangleleft gh)_u$



map sends $A \in \mathfrak{g} \to A^* \in \mathfrak{X}(P)$ | $A^*_u f = \frac{d}{dt} f(u \triangleleft e^{tA}) \Big|_{t=0}$

$f : M \to N$

$df = f_*$

prop 5.1 $\quad \boxed{A^* \triangleleft g = (A \triangleleft g)^*}$

$(Ad_{(g^{-1})}(A))^* = (g^{-1} A g)^* = (dR_{g^{-1}})_{u \triangleleft g} (A^*_{u \triangleleft g})$

A^* I would call X_A

---

if P principal G-bundle $\pi : P \to M$

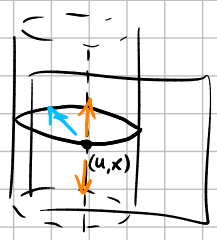
then $M \cong P/G$ quotient by right action

$$\overset{\shortparallel}{\{[u] \mid u \in P\}} \qquad [u] = \{u \triangleleft g \mid g \in G\}$$

<u>idea:</u> bundle with fibre F and base M: $P \times F / G$

<u>example of vertical vector</u>

$E = S^1 \times \mathbb{R}$



fibre at (u,x) : $\{(u,y) \mid y \in \mathbb{R}\}$

at (u,x) we have tangent space $T_{(u,x)}E$

$E = \mathbb{R} \times S^1$

$X : U \underset{open}{\subseteq} P \to TP$ vector field

X is vertical if its integral curve with initial condition $\gamma(0) = u$ is contained in $\{u \triangleleft g \mid g \in G\}$ (fibre containing u)

---

vertical vector field are exactly the A^* with $A \in \mathfrak{g}$

- # of vertical directions is $\dim(F)$ (locally: $M \times F$)

$\pi : E \to M$ bundle with fibre F

$\psi : \pi^{-1}(V) \to V \times F$ $\quad U \cap V \neq \phi$

$p \in \pi^{-1}(U)$ $\quad \phi : \pi^{-1}(U) \to U \times F$

$d(\phi \circ \psi^{-1})$ send vertical to vertical (sends non vertical to zero)

$d\phi_p : T_pE \to T_{\phi(p)}(U \times F) \cong T_u U \oplus \underbrace{T_f F}$ $\quad \phi(p) = (u, f)$

$v \in T_pE$ vertical if $d\phi_p(v)$ vertical

---

want to see that A^*_u is vertical and that $\dim\{A^*_u \mid A \in \mathfrak{g}\} = \dim G = \dim(\mathfrak{g})$

every vertical vector looks like A^*

$\forall u \in P$ map $A \in \mathfrak{g} \mapsto A^*_u \in T_uP$ is <u>injective</u>

$\to \ker = \{0\}$

follows from the fact that action is <u>free</u> $\to$ ($\exists u$ s.t $u \triangleleft g = u$ then $g = e$)

if $A \neq 0$ then $A^*_u \neq 0$ $\forall u \in P$

$$A^*_u f = \frac{d}{dt} f(u \triangleleft \underbrace{e^{tA}}_{\neq e})\Big|_{t=0}$$

prove this

$V_u = \{A^*_u \mid A \in \mathfrak{g}\} \subseteq T_uP$ vertical subspace

$T_uP = V_u \oplus H_u$ horizontal vectors

only requirement: action of G on vector fields sends vertical to vertical and <u>horizontal to horizontal</u>

<u>Connection</u>: smooth choice of H_u at each point $u \in P$ s.t.

$dR_g(H_u) = H_{g \triangleleft u}$

To each connection we can associate a $\mathfrak{g}$-valued 1-form $\omega : P \mapsto T^*P \otimes \mathfrak{g}$

$u \mapsto (u, \omega_u)$

$\omega_u \in T^*_uP \otimes \mathfrak{g}$

if X vector field $\omega_u(X_u) \in \mathfrak{g}$

$X(u) = (u, X_u)$ $\quad X_u \in T_uP$

look at coordinates: $\{E_\mu\}$ basis for vector fields

$$X = X^\mu E_\mu$$

$\{\varepsilon^\mu\}$ dual basis for 1-forms

$$\omega = \omega_\mu \varepsilon^\mu$$

in coords $\varepsilon^\mu = dx^\mu$ $\omega = \omega_\mu dx^\mu$

for regular 1-form $\omega_\mu : P \to \mathbb{R}$

for $\mathfrak{g}$-valued 1-form $\omega_\mu : P \to \mathfrak{g}$

$\omega_u(X_u) =$ the unique $A \in \mathfrak{g}$ s.t. $A^*_u = \mathrm{proj}_{V_u}(X_u) \in V_u$] def of ω

$X_u \in T_u P = V_u \oplus H_u$

Prop 1.1

(a') $\omega_u(A^*_u) = A \quad \forall A \in \mathfrak{g}$

(b') $R^*_a \omega = \mathrm{ad}(a^{-1})\omega = a^{-1}\omega a \quad \forall a \in G$ pullback

$\Rightarrow (R^*_a \omega)_u(v) = a^{-1}\omega_u(v)\, a$

(c') if X_u horizontal $\omega_u(X_u) = 0$

$$[\phi(t,x), \pi(t,x')] = i\hbar\, \delta^3(x, x')$$ ← flat spacetime

$\pi = \frac{\partial \phi}{\partial t}$

$\mathbb{R} \times M$